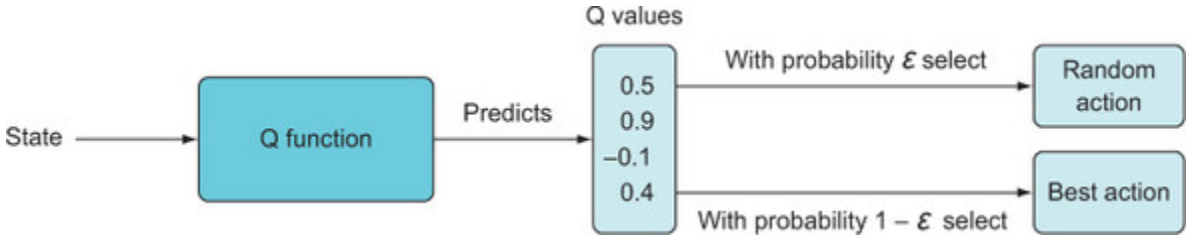
network to decide what action to take using some action selection procedure, such as a simple epsilon-greedy approach or a softmax selection policy. In this chapter we'll use the epsilon-greedy approach (figure 3.6) as DeepMind did, and instead of using a static ε value like we did in the last chapter, we will initialize it to a large value (i.e., 1, so we'll start with a completely random selection of actions) and we will slowly decrement it so that after a certain number of iterations, the ε value will rest at some small value. In this way, we will allow the algorithm to explore and learn a lot in the beginning, but then it will settle into maximizing rewards by exploiting what it has learned. Hopefully we will set the decrementing process so that it will not underexplore or overexplore, but that will have to be tested empirically.

Figure 3.6. In an epsilon-greedy action selection method, we set the epsilon parameter to some value, e.g., 0.1, and with that probability we will randomly select an action (completely ignoring the predicted Q values) or with probability 1 – epsilon = 0.9, we will select the action associated with the highest predicted Q value. An additional helpful technique is to start with a high epsilon value, such as 1, and then slowly decrement it over the training iterations.



We have the output layer figured out—now to tackle the rest. In this chapter, we will construct a network of just three layers with widths of 164 (input layer), 150 (hidden layer), 4 (the output layer you already saw). You are welcome and encouraged to add more hidden layers or to play with the size of the hidden layer—you will likely be able to achieve better results with a deeper network. We chose to implement a fairly shallow network here so that you can train the model with your own CPU (it takes our MacBook Air 1.7 GHz Intel Core i7, with 8 GB of RAM, only a few minutes to train).

We already discussed why the output layer is of width 4, but we haven't talked about the input layer yet. Before we do that, though, we need to introduce the Gridworld game engine we will be using. We developed a